Flight Fare Exploratory Data Analysis (EDA)

A Python data analysis project that explores flight-fare data using Pandas, NumPy, Matplotlib, and Seaborn.

Project Overview

This notebook performs exploratory data analysis (EDA) on a flight fare dataset. It inspects the dataset, summarizes numerical columns, checks missing values, creates journey date features, and visualizes patterns in prices, airlines, routes, stops, and flight duration.

Objectives

- Understand the dataset structure, column data types, and non-null values.
- Review descriptive statistics for numerical features.
- Identify columns containing missing values.
- Convert Date_of_Journey to a date and extract journey day, month, and year.
- Visualize flight-price and journey-date distributions.
- Compare airline, source, destination, and total stops with flight prices.
- Convert flight duration text into total minutes and examine its relationship with price.

Tools and Libraries

Python 3, Pandas, NumPy, Matplotlib, and Seaborn.

Notebook Contents

- Dataset loading and DataFrame inspection.
- Descriptive statistics and missing-value checks.
- Date feature engineering.
- Histograms and count plots for numerical/date and categorical features.
- Box plots comparing categories with Price.
- Duration conversion and a scatter plot of duration versus price.

Dataset Requirement

The notebook expects a CSV file named `flight_fare_dataset.csv` in the same working directory. If your dataset has a different filename, update the `pd.read_csv()` line. The notebook includes a small dummy-data fallback when the file is missing; charts produced from that fallback are only for demonstration and are not real dataset findings.

How to Run

Google Colab: Upload `eda1.ipynb` and the flight fare CSV file to Colab, then run the notebook cells from top to bottom.

Local computer: Install Python 3 and the required libraries with `pip install pandas numpy matplotlib seaborn jupyter`. Place the CSV file beside the notebook and open the notebook in Jupyter.

Project Structure

```
flight-fare-EDA/
■■■ eda1.ipynb
■■■ flight_fare_dataset.csv
■■■ README.pdf
```

Learning Outcomes

This project practices data loading, data inspection, descriptive statistics, missing-value analysis, feature engineering, and visual exploration of relationships between flight attributes and prices.

Possible Improvements

Future improvements could include stronger data cleaning, outlier analysis, more detailed insights, and a predictive model for flight fares.

Created as a data analysis portfolio project.